

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
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Operations Research

Exam: IN0024 / Midterm

Date: Friday 6th June, 2025

Examiner: Prof. Dr. Martin Bichler

Time: 15:30 – 16:30

	P 1	P 2	P 3	P 4	P 5
I					
II					

Working instructions

- This exam consists of **12 pages** with a total of **5 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 60 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
 - one **ruler/geo triangle**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Answer the exam questions in English or German.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Problem 1 Potatoes Supply (11 credits)

You are the owner of a restaurant chain with $B = \{1, \dots, 5\}$ branches and want to formulate a linear program to model the optimal supply of potatoes for the next month. There are $S = \{1, \dots, 10\}$ suppliers from which you can procure potatoes. The cost of procuring 1 kg of potatoes from supplier s is $c_s \in \mathbb{R}_{>0}$.

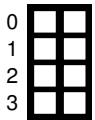
Let $x_{bs} \in \mathbb{R}_{\geq 0}$ denote the kilograms of potatoes procured from supplier s for branch b , for all $b \in B, s \in S$.

Formulate the following constraints using only linear (in)equalities.



a) Each branch $b \in B$ has a demand $d_b \in \mathbb{R}_{\geq 0}$ of potatoes which must be exactly met.

$$\sum_{s \in S} x_{bs} = d_b \quad \forall b \in B$$



b) Each supplier $s \in S$ has to supply at least $m_s \in \mathbb{R}_{\geq 0}$ and at most $M_s \in \mathbb{R}_{\geq 0}$ kilograms in total.

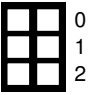
$$m_s \leq \sum_{b \in B} x_{bs} \leq M_s \quad \forall s \in S$$



c) Assume that each supplier has an infinite supply capacity, i.e., $M_s = \infty \quad \forall s \in S$. Justify whether there always exists a solution satisfying the constraints defined in parts a) and b).

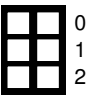
No. If the total demand is less than the total minimum required by suppliers, i.e., $\sum_{b \in B} d_b < \sum_{s \in S} m_s$, then no feasible solution exists.

d) At least twice as many potatoes should be procured from supplier 1 compared to supplier 2.



$$\sum_{b \in B} x_{b1} \geq 2 \cdot \sum_{b \in B} x_{b2}$$

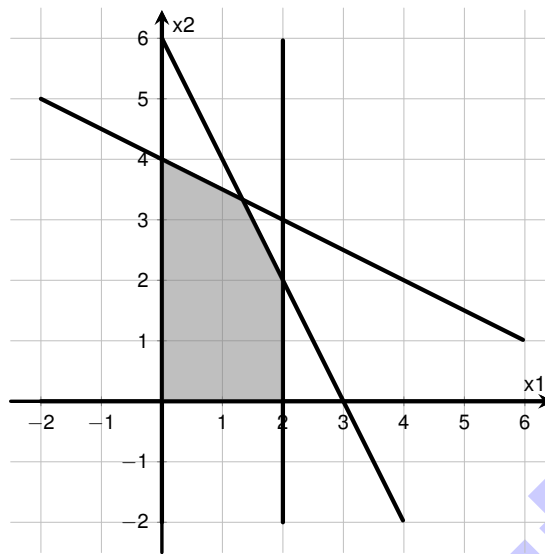
e) Formulate an objective function that minimizes total procurement costs.



$$\min \sum_{s \in S} \sum_{b \in B} c_s x_{bs}$$

Problem 2 Graphical Solution (11 credits)

Given is the following feasible region (shaded in gray) of a linear problem.



a) Specify linear constraints that define this feasible region.

Possible inequalities:

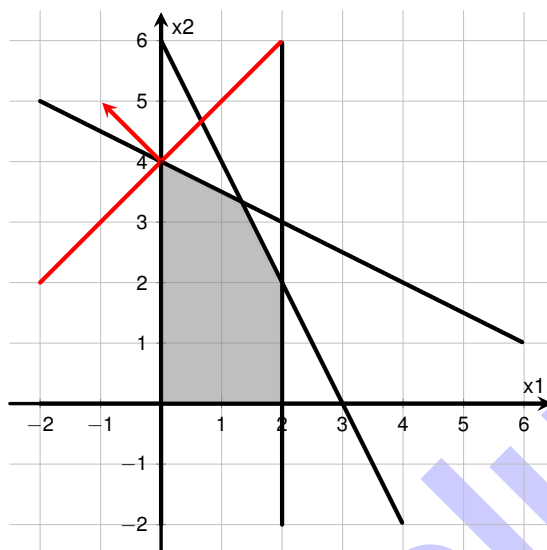
- $0.5x_1 + x_2 \leq 4$
- $2x_1 + x_2 \leq 6$
- $x_1 \leq 2$
- $x_1, x_2 \geq 0$

	0
	1
	2
	3

b) Specify an objective vector c such that $(0, 4)$ is the only optimal solution of the problem $\max c^T x$ subject to the constraints from part a). Draw your objective function.

Note: You may also use the figure's copy below to draw your objective function. Clearly mark which one you want to be graded!

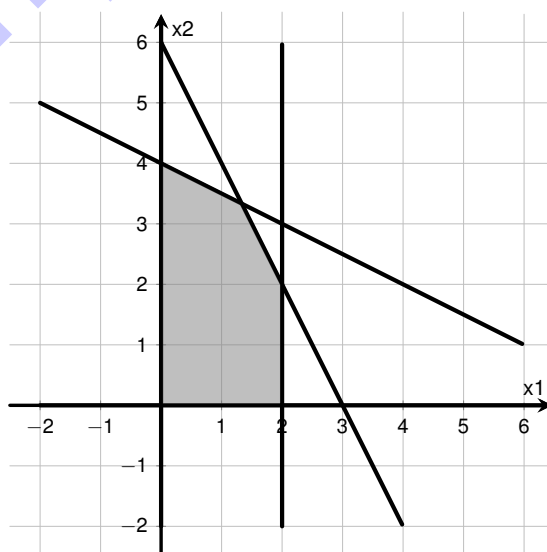
Possible solution: $c = (-1, 1)^T$



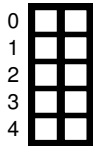
c) Consider the objective vector $c' = (-1, \beta)^T$. For which choices of $\beta \in \mathbb{R}$ is the set of optimal solutions more than a single point?

	0
	1

The only possible solution is $\beta = 0$.



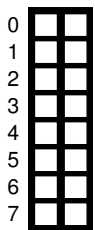
Problem 3 Simplex Tableaus (15 credits)



a) Transform the following LP into canonical form.

$$\begin{array}{ll} \text{Max} & x_1 + 2x_2 \\ \text{s.t.} & -2x_1 + x_2 \geq -2 \\ & -x_1 + x_2 \leq 3 \\ & x_1 + x_2 \leq 7 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

$$\begin{array}{ll} \text{Max} & x_1 + 2x_2 = z \\ \text{s.t.} & 2x_1 - x_2 + x_3 = 2 \\ & -x_1 + x_2 + x_4 = 3 \\ & x_1 + x_2 + x_5 = 7 \\ & x \in \mathbb{R}_{\geq 0}^5 \end{array}$$



b) Perform one iteration of the simplex algorithm. State the incoming and outgoing basic variables. You can use the simplex tableaus at the end of the exercise. Please indicate clearly if you do so.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-3			2		6
x_3	1		1	1		5
x_2	-1	1		1		3
x_5	2			-1	1	4

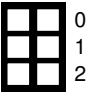
Is the resulting tableau optimal?

Basis swap $x_5 \leftrightarrow x_1$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z				1/2	3/2	12
x_3			1	3/2	-1/2	3
x_2		1		1/2	1/2	5
x_1	1			-1/2	1/2	2

Since there are only positive entries in the objective row, the stopping criterium is met and the tableau is optimal.

c) For the following LP in tableau notation, provide the largest possible ranges for α and β such that one can read directly from the simplex tableau that the underlying LP is unbounded. Justify your answer.



Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	α			2		6
x_3	-1		1	1		5
x_2	-1	1		1		3
x_5	β			1	-1	-4

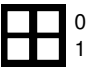
$\alpha < 0$: Implies that x_1 is the incoming basic variable, $\beta \geq 0$: Note that the tableau is not in canonical form. The last row has to be multiplied by -1 . If all coefficients are negative in the x_1 columns, then the LP is unbounded. Hence, $-\beta \leq 0 \Leftrightarrow \beta \geq 0$.

d) Can the same vertex be visited more than once during the simplex algorithm? Justify your answer.



Yes, if the problem is degenerate.

e) Can a vertex be revisited after a different vertex was visited? Justify your answer.



No, since the objective strictly increases when visiting a different vertex.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Base	Coefficients					Res.

Problem 4 Revised Simplex (12 credits)

Use the revised simplex method to perform **one** iteration of the following linear program. Begin with the initial basic feasible solution given by the basis variable x_2 and the slack variables s_2 and s_3 . State which variables enter and exit the basis or why the algorithm stops, if it does.

$$\begin{aligned} \min \quad & 3x_1 - 2x_2 - 2x_3 \\ \text{s.t.} \quad & -4x_1 - 5x_2 + x_3 \geq -20 \\ & 2x_1 + x_2 - 4x_3 \leq 8 \\ & 5x_1 + 7x_2 \leq 35 \\ & x_1, x_2 \geq 0 \\ & x_3 \leq 0 \end{aligned}$$

Hint 1: You can use in iteration 1:

$$B^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 5 & 0 \\ -7 & 0 & 5 \end{pmatrix}$$

Hint 2: Solution after iteration 1:

$$x_1 = 0, x_2 = 4, x'_3 = 0, s_1 = 0, s_2 = 4, s_3 = 7, Z = -8.$$

We first bring the LP into canonical form by negating the first constraint, introducing $x'_3 = -x_3$ and adding slack variables s_1, s_2, s_3 . (Note: Normally, the objective function would also be negated to yield a maximisation problem; however, this step can be skipped if we subsequently check that $c'_N \leq 0$ rather than $c'_N \geq 0$.)

$$\begin{aligned} \min \quad & 3x_1 - 2x_2 + 2x'_3 \\ \text{s.t.} \quad & 4x_1 + 5x_2 + x'_3 + s_1 = 20 \\ & 2x_1 + x_2 + 4x'_3 + s_2 = 8 \\ & 5x_1 + 7x_2 + s_3 = 35 \\ & x_1, x_2, x'_3, s_1, s_2, s_3 \geq 0 \end{aligned}$$

Now $x_B = (x_2, s_2, s_3)^T$, $x_N = (x_1, x'_3, s_1)^T$.

$$B = \begin{pmatrix} 5 & 0 & 0 \\ 1 & 1 & 0 \\ 7 & 0 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 4 & 0 \\ 5 & 0 & 0 \end{pmatrix}, \quad c_B = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}, \quad c_N = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$$

(When the objective function is negated, the signs of c_B and c_N are reversed.)

$$B^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 0 & 0 \\ -1 & 5 & 0 \\ -7 & 0 & 5 \end{pmatrix}, \quad N' = B^{-1}N = \frac{1}{5} \begin{pmatrix} 4 & 1 & 1 \\ 6 & 19 & -1 \\ -3 & -7 & -7 \end{pmatrix}$$

$$\begin{aligned}
 c'_N &= -(c_N - N'^T c_B) \\
 &= - \left(\begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 8 \\ 2 \\ 2 \end{pmatrix} \right) = -\frac{1}{5} \begin{pmatrix} 23 \\ 12 \\ 2 \end{pmatrix}
 \end{aligned}$$

For minimization, we search for the largest positive reduced cost. Since all $c'_N \leq 0$, optimality is reached. If the objective function had been negated, the reduced costs c'_N would be entirely positive, and there would be no negative one to continue.

Sample Solution

Problem 5 Sensitivity Analysis (11 credits)

A café wants to introduce two new items into their breakfast menu, waffles and pancakes, to make use of the resources left in their inventory, milk and flour. They gathered all required information in the table below and want to find prices for the new items, c_1 and c_2 Euros respectively.

Product	Milk (l)	Flour (kg)	Price (€)
Waffle	1	2	c_1
Pancake	1	1	c_2
Capacity	5	8	

They gave these information to ChatGPT and asked for a solution that would use all of their resources. ChatGPT returned the following linear program and the simplex tableau.

$$\begin{aligned}
 \max \quad & c_1 x_1 + c_2 x_2 \\
 \text{s.t.} \quad & x_1 + x_2 \leq 5, \\
 & 2x_1 + x_2 \leq 8, \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

Base	Coefficients				Res.
	x_1	x_2	s_1	s_2	
z	0	0	$2c_2 - c_1$	$c_1 - c_2$	$3c_1 + 2c_2$
x_1	1	0	-1	1	3
x_2	0	1	2	-1	2

Answer the following short questions to check if the answer from ChatGPT is accurate.



a) What do s_1 and s_2 stand for?

The variables s_1 and s_2 are the slack variables. They stand for the respective unused resources.



b) What are the basic variables? Why did ChatGPT choose such a basis?

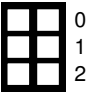
The basic variables are x_1 and x_2 . ChatGPT chose such a solution since the user asked for a solution that uses all their resources, which can only happen if the slack variables are non-basic, and x_1 and x_2 are basic variables.



c) How many of each resource is left if the proposed solution is undertaken?

In the proposed solution, there are no resources left since the corresponding slack variables are non-basic.

d) Assume that the values in the simplex tableau are correct. Find the set of inequalities that c_1 and c_2 should satisfy so that the current basis would be optimal.



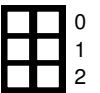
For optimality, we should have non-negative reduced costs:

$$2c_2 - c_1 \geq 0, \quad \& \quad c_1 - c_2 \geq 0.$$

For the following exercises assume $c_1 = 3$ and $c_2 = 2$.

e) Should the firm purchase extra units of milk at the price of 0.80 Euros per liter? How does the optimum objective change if one extra liter is bought?

Hint: The slack variable for milk is s_1 .



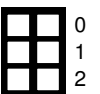
Given $c_1 = 3$ and $c_2 = 2$, the shadow price of milk is $2c_2 - c_1 = 1$. The purchase cost is €0.80/liter, so the net gain is

$$1 - 0.80 = 0.20 \quad \text{€per liter.}$$

Since the net gain > 0 , buying an extra unit is profitable. The objective increases by 0.20 €.

f) Find the allowable range for the capacity of milk over which the current basis $\{x_1, x_2\}$ remains feasible.

Hint: The slack variable for milk is s_1 .



Allowable range for the capacity of milk: Let the capacity change to $5 + \Delta_m$. We can use the column under the slack variable s_1 to find the changes in the RHS. For feasibility, the RHS values should remain non-negative:

$$3 - \Delta_m \geq 0 \Rightarrow \Delta_m \leq 3, \quad 2 + 2\Delta_m \geq 0 \Rightarrow \Delta_m \geq -1.$$

Hence, the current basis remains valid for $\Delta_m \in [-1, 3]$.

g) Explain why the current shadow price of milk becomes invalid if its capacity moves outside that range.



If the capacity of milk falls below 4 or rises above 8, one of the basic variables (x_1 or x_2) becomes negative. The current basis is no longer feasible, so the tableau—and thus the shadow-prices—must be recomputed under a new basis.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

